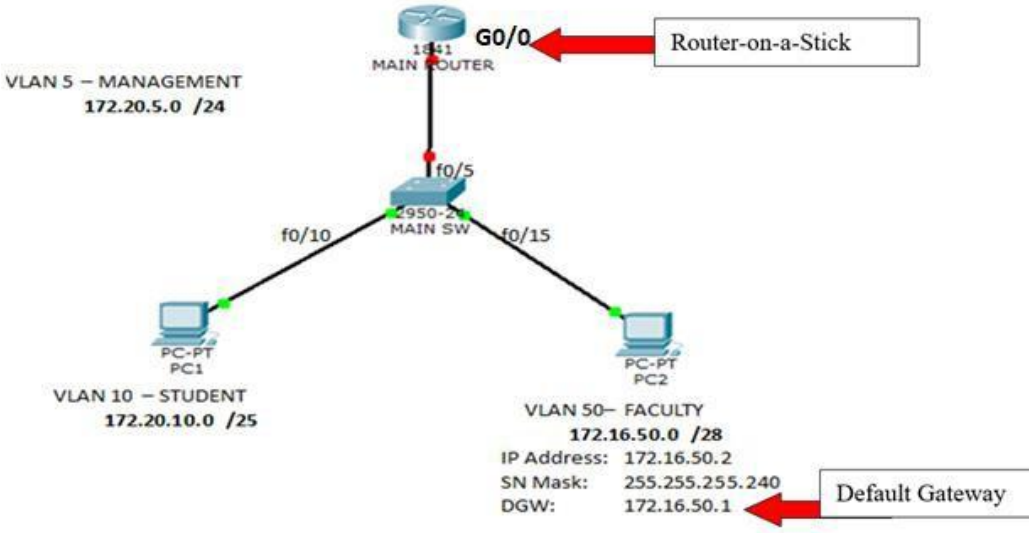
Net Admin 2 – Practical MidTerm Exam (100 points)

For this exam, you will take the given Packet Tracer file, called "Practical Midterm", and follow the steps given below. You may use any resource found on D2L or Netacad, including labs, personal notes or previous Zoom sessions to assist you. You may not ask for help to complete the exam or have someone else do the work for you. If you need some clarification on the exam, please email me. When you are finished, please upload the completed Packet Tracer File and the network chart to the assignment box in D2L called "Pratical MidTerm."

Step 1: Open the file, fill out the chart

A. Open up the Packet Tracer file called "Practical MidTerm", in the Module 07 (MidTerm) folder on D2L. The topology and all of the wiring has been done for you.



B. Using the given Networks, fill out the chart below: **(10 points)** -**Submit a screenshot to D2L.**

Network Name and VLAN	Network Address and SNM	First Valid IP in that VLAN/Subnet	Last Valid IP in that VLAN/Subnet	Broadcast Address	Subnet Mask (SNM)
Management and Native VLAN 5	172.20.5.0 /24	172.20.5.1 /24	172.20.5.254 /24	172.20.5.255 /24	255.255.255.0
Student Network VLAN 10	172.20.10.0 /25	172.20.10.1 /25	172.20.10.126 /25	172.20.10.127 /25	255.255.255.128
Faculty Network VLAN 50	172.16.50.0 /28	172.16.50.1 /28	172.16.50.14 /28	172.16.50.15 /28	255.255.255.240

Step 2: Configure the Switch

A. Apply a hostname and passwords using the following parameters. **(10 points)**

Parameter	Switch	Points
Host name:	Test-Switch.	2
Enable password:	Rams	3
Console password:	Cardinals	2
Telnet (VTY port) password:	Blues	3

B. Create and name the VLANs for the Student, Faculty, and Management Subnets. **(10 points)**

Parameter	Switch	Points
Create VLANS	VLAN 5, 10, & 50	5
Name VLANS	Management, Student, & Faculty – (See the above diagram for more info)	5

C. Based on the port that the PCs and router are plugged into, configure each port with the correct mode and assign it into the correct VLAN. **(20 points)**

Parameter	Switch	Points
Fa0/10 & fa0/15	Configure as access ports	5
Fa0/10	Add to Vlan 10	5
Fa0/15	Add to Vlan 50	5
Fa0/5 (plugged into router)	Configure as a trunk port	5

D. Assign an IP address and subnet mask to the Management/Native VLAN. **(4 points)**

Parameter	Switch	Points
Management- VLAN 5	Assigning an IP address 172.20.5.1	2
	Assigning a Subnet Mask 255.255.255.0	2

Step 3: Configure the Router

A. Apply a hostname and passwords using the following parameters. **(10 points)**

Parameter	Router	Points
Host name:	R1	2
Enable password:	Cisco	3
Console password:	Minneapolis	2
Telnet (VTY port) password:	Yellow	3

B. Configure a sub-interface on the correct physical interface for the VLANs and then turn on the physical interface. **(20 points)**

What to do	Router	Points
Turn on the physical interface	G0/0	5
Configure a sub-interface on the VLAN 5. Then assign a static IP and SUBNET MASK.	Choose the correct sub-interface	5
Configure a sub-interface on the VLAN 10. Then assign a static IP and SUBNET MASK.	Choose the correct sub-interface	5
Configure a sub-interface on the VLAN 50. Then assign a static IP and SUBNET MASK.	Choose the correct sub-interface	5

Step 4: Configure the PCs (10 points)

Assign a static IP, SUBNET MASK and Default Gateway on the Student and Faculty PC's from the correct subnets.

What to do	Computer	Points
Assign a static IP, SUBNET MASK, and Default Gateway.	PC1	5
Assign a static IP, SUBNET MASK, and Default Gateway.	PC2	5

Step 5: Test your Network (6 points)

Test your Network	Computer	Points
Are the PC's able to ping each other?	PC1 –> PC2	3
Are the PC's able to ping their Default Gateway?	PC1 –> R1 PC2 –> R1	3